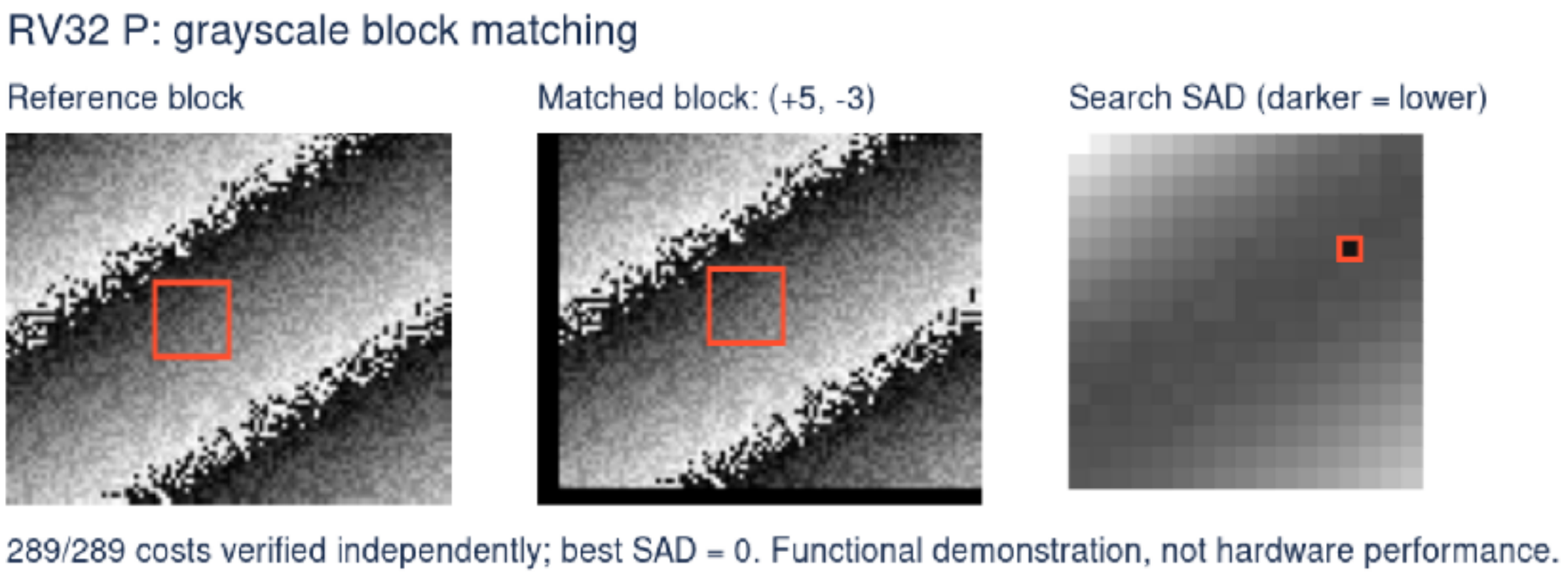
Image block matching, now on RV32 P

Add absolute pixel differences. The lowest SAD identifies the match.



Four pixels per packed operation

The image kernel is C. No inline assembly is required.

```
#include <riscv_packed_simd.h>

uint64_t sum = 0;
size_t x = 0;
for (; width - x >= 4; x += 4)
  {
    uint8x4_t av = __riscv_pld_u8x4 ((uint8_t *) (a + x));
    uint8x4_t bv = __riscv_pld_u8x4 ((uint8_t *) (b + x));
    sum += __riscv_pabdsumu_u8x4_u32 (av, bv);
  }
```

LOAD

Four pixels from each image

REDUCE

Four absolute byte differences

ACCUMULATE

Add to a 64-bit total

Generated packed SAD: pabdsumu.b / pabdsumau.b

Full kernel: row strides, alignment handling and scalar tails. RV64 uses eight byte lanes.

An integer-core demo, with clear assumptions

P uses the integer registers; this kernel needs neither floating point nor RVV.

RV32 KERNEL

RV32IMC + Zba/Zbb + P

P draft 0.6 · ILP32 soft-float ABI

The Linux demo runtime also needs A.
The SAD kernel does not.

HARDWARE MODE

Direct word loads.
Requires supported misaligned accesses.

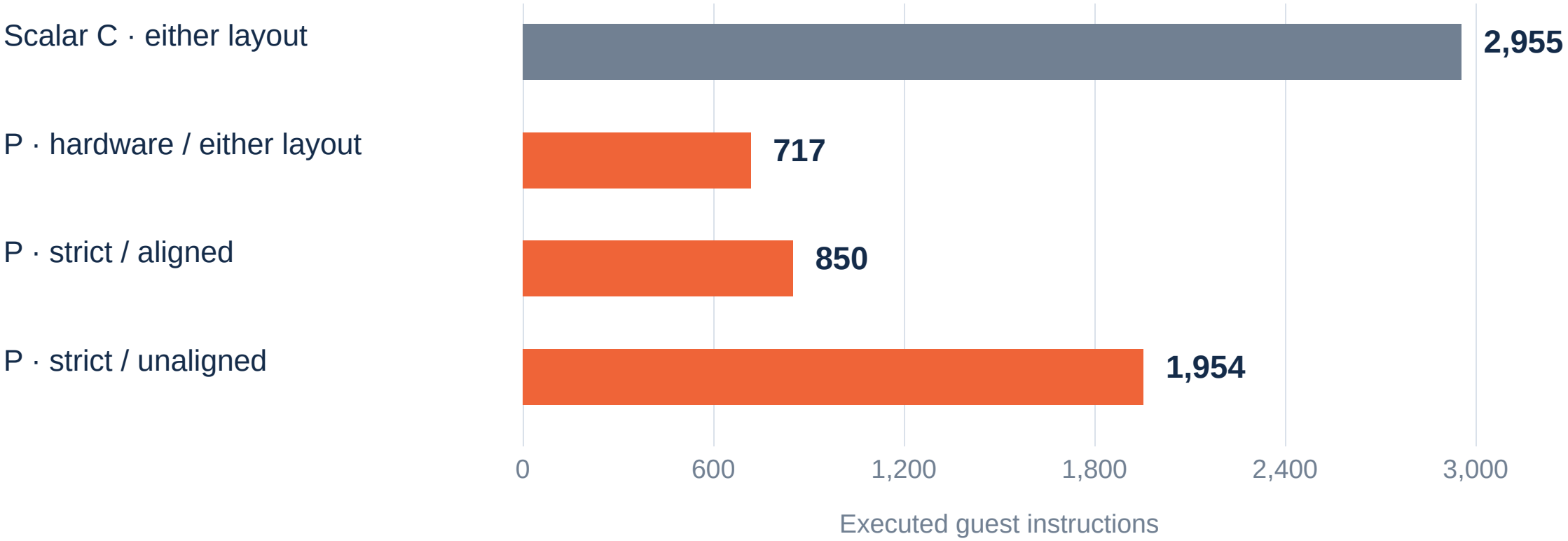
STRICT MODE

Check alignment.
Use byte loads and packing when needed.

Misaligned-load support is a target assumption, not a P requirement.

RV32: 76% fewer executed instructions

One 16 × 16 kernel call, including return · -O3 · lower is better



Identical logical pixels; SAD 22,364. Same -mtune=size; no LTO or auto-vectorization.
Equal aligned/unaligned counts do not imply equal hardware memory latency.

Instruction savings have a footprint

RV32 hardware mode · kernel-only measurements · no application-size claim

Metric	Scalar · -O3	P · -O3	Scalar · -Os	P · -Os
Guest instructions	2,955	**717**	3,334	**845**
Kernel text, bytes	108	**204**	102	**132**
Static frame, bytes	0	**32**	16	**16**

P at -Os: text 204 → 132 bytes; instructions 717 → 845.

Strict P at -O3: 400 text bytes; 48-byte static frame.

GCC -fstack-usage reports static kernel frames, not maximum whole-program stack use.

A demo you can reproduce

C kernels, included tools, independently checked image-search costs.

12	82,640	289/289
configurations pass	C checks per build	image costs verified

RV32 / RV64 × hardware / strict alignment × O2 / O3 / Os

```
XLEN=32 ALIGNMENT=hardware bash scripts/present.sh
```

Hardware speedup, memory latency and energy need a compatible board or cycle model.

github.com/pz9115/RVP-Example